

ERROR GEOMETRY AND SUBSPACE PROJECTION FOR NEURAL AC POWER FLOW

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ABSTRACT

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Recent neural power-flow solvers, including emerging foundation models, achieve accurate voltage predictions, yet such accuracy does not necessarily imply physically consistent solutions. Even small voltage errors can yield large residuals from the AC power-balance. We study this accuracy–consistency gap across PIGNN, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus GBnetwork scenarios, with cross-grid evaluation of GridSFM over 31 systems. Using an SVD basis fitted to training AC power-flow solutions, we find that neural prediction errors contain substantial components outside the dominant solution subspace. To address this mismatch, calibrated solution-subspace projection (CSP) removes off-subspace prediction components, substantially reducing mean power-balance residual from 0.3159 to 0.0550 p.u. for PIGNN-GC and from 0.2687 to 0.0514 p.u. for GridSFM. These results highlight output-error geometry as an important factor in physics-consistent neural AC power flow.

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Index Terms— AC power flow, neural surrogate, physics violation, error geometry, calibration, distribution shift

1. INTRODUCTION

Modern power-system operation requires repeated AC power-flow (AC-PF) evaluations across increasingly diverse operating conditions driven by renewable generation, electrification, and changing demand [1, 2]. AC-PF maps network conditions and power injections to bus voltages and flows serving as a fundamental nonlinear primitive in power-system analysis [3]. Repeated evaluations arise in time-series and contingency studies [4], while the same AC network equations underlie optimization and state-estimation tasks [5].

The computational demands of repeated grid analysis have motivated learned surrogates that amortize the cost of conventional solvers over many operating points [6]. Learned grid solvers span different levels of constraint enforcement: direct regression prioritizes fast inference [7], physics- or constraint-aware objectives penalize violations during training [8, 9], while iterative correction and feasibility-seeking mechanisms refine predictions more directly toward physically consistent [10, 11] or feasible states [12, 13]. More recent cross-grid and foundation-model efforts—including GridSFM [14], gridfm-graphkit and GENCO [15, 16], and LUMINA [17]—extend

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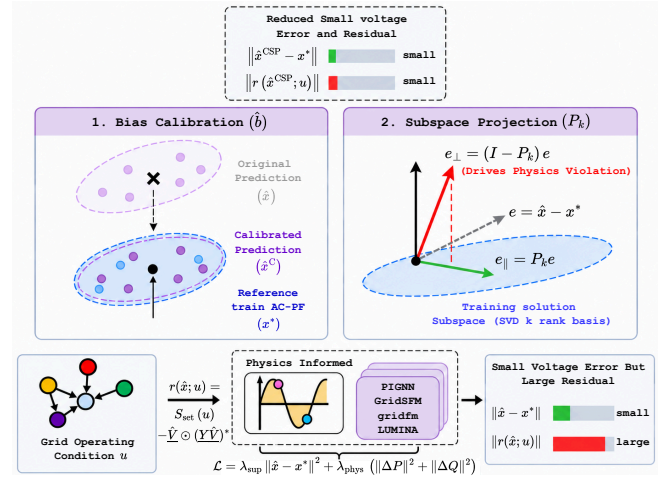


Fig. 1. Neural AC-PF surrogates can achieve small voltage errors while retaining large physics violations. Bias calibration and solution-subspace projection jointly suppress systematic and off-subspace errors, improving both voltage accuracy and AC power-balance consistency.

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this direction with global context and rich heterogeneous, topology-aware representations designed to operate across diverse grids and operating conditions.

Despite these advances, accurate voltage prediction does not necessarily imply AC-consistent output [18]. Physics-informed objectives can substantially reduce power-flow residuals [19, 18], but residual penalties remain soft training objectives and do not by themselves enforce exact satisfaction of the nonlinear AC equations. Consequently, small voltage-state errors can coexist with non-negligible power-balance violations. This raises a question not addressed by state accuracy alone: what properties of the remaining voltage error determine how strongly it is translated into AC power-balance residuals?

We hypothesize that the missing factor is the *direction* of the remaining voltage error: errors of similar magnitude may produce very different power-balance residuals depending on how they are oriented across buses. Goodwin et al. [20] characterized the differential geometry of the AC-PF solution manifold and assessed the quality of its linear approximations. For neural surrogates, we instead study error geometry empirically, asking whether the orientation of voltage errors relative to a linear subspace estimated from training solutions

helps explain the remaining physical inconsistency.

We evaluate this hypothesis across PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus GBnetwork scenarios, with additional cross-grid evaluation of GridSFM over 31 systems of diverse sizes. Across all four models, train-only calibration improves voltage accuracy while leaving persistent power-balance residuals, showing that persistent bias alone does not explain the accuracy–consistency gap. We therefore construct an SVD basis exclusively from training AC power-flow solutions and examine the orientation of the remaining voltage errors relative to the resulting training-supported solution subspace. **Unlike Park et al., who use a PCA-based subspace to compress and reconstruct AC-OPF outputs during learning [21], we use the training-solution subspace to analyze surrogate-error geometry and to correct frozen AC-PF predictions post hoc.**

We find that substantial prediction error can lie outside the dominant training-supported solution directions and that suppressing this off-subspace component markedly reduces AC power-balance violations. Motivated by this observation, we introduce *calibrated solution-subspace projection* (CSP), which first removes train-estimated bias and then suppresses prediction components outside the dominant training-supported directions. CSP improves voltage-magnitude accuracy and AC power balance simultaneously, while cross-grid and distribution-shift experiments show that its benefits extend across diverse systems but depend on how well the training-derived subspace represents the operating distribution. Together, these results identify voltage-error geometry across buses as an important link between state accuracy and physical consistency.

Code: github.com/Kimchangheon/neural-acpf-error-geometry

2. ERROR GEOMETRY AND CALIBRATED SUBSPACE PROJECTION

2.1. AC power-balance consistency and train-only calibration

We consider an N -bus grid with voltage state $x = [V, \theta] \in \mathbb{R}^{2N}$, where $V = V \odot e^{j\theta}$. Given an operating condition u , a neural surrogate predicts $\hat{x} = f_\phi(u)$, from which the AC active and reactive powers (P, Q) are recomputed using the network admittance model. For setpoints $(P^{\text{set}}, Q^{\text{set}})$, define

$$r(\hat{x}; u) = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P^{\text{set}} - P \\ Q^{\text{set}} - Q \end{bmatrix}, \quad \text{PB} = \sqrt{\Delta P^2 + \Delta Q^2}.$$

AC power-balance consistency is assessed by PB (Power Balance), evaluated bus-wise under the standard slack/PV/PQ conventions; **Mean PB and Max PB are reported over buses and scenarios.** Before geometric analysis, we remove train-only post-training bias:

$$\hat{b} = \mathbb{E}_{s \in \text{tr}} [\hat{x}_s - x_s^*], \quad \hat{x}^C = \hat{x} - \hat{b},$$

with $\hat{b}$ frozen at test time. Angle offsets use circular differences after gauge alignment; calibration removes bias without constraining residual error directions.

2.2. Error geometry in a training-derived solution subspace

To characterize the remaining error, we construct a rank- k -per-channel basis from Newton–Raphson solutions in the training set. Let X_V and X_θ denote the centered training matrices of voltage magnitudes and gauge-aligned angles, respectively. We compute separate SVDs

$$X_V = W_V \Sigma_V U_V^\top, \quad X_\theta = W_\theta \Sigma_\theta U_\theta^\top, \quad P_k = W_k W_k^\top, \quad (1)$$

where $W_k = \text{diag}(W_{V,k}, W_{\theta,k})$ contains the leading k directions from each SVD; k is capped by the available components on smaller systems.

These subspaces capture dominant training-solution variation and define our error coordinates. For $e = \hat{x} - x^*$, we decompose

$$e^\parallel = P_k e, \quad e^\perp = (I - P_k) e, \quad (2)$$

where $e^\parallel$ lies along the dominant training-solution directions and $e^\perp$ captures error outside them.

To relate this error decomposition to physical consistency, we use a first-order Taylor expansion of the AC residual around a converged power-flow solution x^* :

$$r(x^* + e; u) = \underbrace{r(x^*; u)}_{=0} + \underbrace{J(x^*; u)e}_{\text{first-order term}} + \underbrace{\mathcal{O}(\|e\|^2)}_{\text{higher-order terms}}, \quad (3)$$

where J is the AC residual Jacobian. Thus, the Jacobian determines how a given voltage error is translated into power-balance residual: errors aligned with directions of larger Jacobian gain can produce substantially larger residuals even when their magnitudes are similar. This motivates testing whether AC power balance changes when the voltage error is rotated toward or away from the training-derived solution subspace while its magnitude is held fixed.

2.3. Calibrated solution-subspace projection (CSP)

This motivates a direct intervention: suppress prediction components outside the dominant training-solution directions while preserving the variation supported by the data. Because systematic bias and off-subspace error are distinct, we first apply post-training calibration and **then project the calibrated prediction:**

$$\hat{x}^{\text{CSP}} = \bar{x}_{\text{tr}} + P_k (\hat{x} - \hat{b} - \bar{x}_{\text{tr}}) \quad (4)$$

where $\hat{b}$ is estimated from the trained model’s predictions on the training set and frozen at test time, while $\bar{x}_{\text{tr}}$ and P_k are obtained solely from the training reference solutions.

The projection removes off-subspace prediction error, but can also discard valid reference variation. To characterize this trade-off, let $e^C = \hat{x}^C - x^*$. **Since P_k is orthogonal,**

$$\|e^{\text{CSP}}\|_2^2 - \|e^C\|_2^2 = \|(I - P_k)(x^* - \bar{x}_{\text{tr}})\|_2^2 - \|(I - P_k)e^C\|_2^2. \quad (5)$$

Thus, CSP improves state accuracy when the removed off-subspace prediction error exceeds the valid reference variation discarded by the projection.

CSP is a data-driven output correction rather than a projection onto the nonlinear AC-feasible set, and therefore does not guarantee exact satisfaction of the AC equations or operational constraints. **Instead, it tests whether suppressing off-subspace prediction components improves voltage accuracy and recomputed AC power-balance residuals.**

3. EXPERIMENTS AND RESULTS

3.1. Experimental setup

Our primary study uses **37,022 converged pandapower Newton–Raphson operating points on the 2224-bus GBnetwork, split 1:1:1 for training/validation/test and varying load, generation, and PV set-points [22].** We evaluate PIGNN-GC, our global-context extension

of PIGNN-Attn-LS [23], GridSFM [14], gridfm-graphkit v0.9.0 [15], and LUMINA [17]; GridSFM is additionally evaluated cross-grid on 31 systems.

All models are trained from scratch using model-specific stable initialization, and CSP is evaluated within each frozen model. gridfm-graphkit uses its released architecture/configuration without a pretrained checkpoint. GridSFM retains the released backbone but is adapted from AC-OPF to AC-PF by fixing generator dispatch and training only for voltage-state prediction with PF-specific residuals. LUMINA likewise retains its released backbone but uses an enhanced AC-PF configuration with known-variable masking, electrical bus features, and stabilized initialization. For all models, electrical inputs, admittance, and evaluation use the same pandapower internal PPC representation [24, 25].

All calibration statistics and solution bases are fitted on training data only. From $k \in \{4, 8, 16, 32, 64\}$, validation Mean PB selects $k^* = 16$ without degrading calibrated voltage RMSE; this rank is then frozen for all test and distribution-shift experiments.

Voltage RMSE alone can be small even when a model largely predicts a static per-bus profile rather than following scenario-dependent voltage variation [14]. To isolate this scenario-dependent response, we center each bus as $\tilde{V}_{is} = V_{is} - \bar{V}_i$ and $\tilde{V}_{is} = \hat{V}_{is} - \bar{V}_i$, and report

$$R_w^2 = \text{corr}^2(\tilde{V}_{is}, \hat{\tilde{V}}_{is}), \quad a_w = \frac{\text{Cov}(\tilde{V}_{is}, \hat{\tilde{V}}_{is})}{\text{Var}(\tilde{V}_{is})}. \quad (6)$$

R_w^2 is the squared correlation between the centered prediction and reference, while a_w measures the magnitude of the predicted variation relative to the true variation.

3.2. Fixed-norm directional intervention.

We vary the orientation of the voltage-magnitude error relative to the training-derived solution subspace while holding its ℓ_2 norm fixed, thereby isolating direction from error magnitude. Starting from the calibrated prediction, we define

$$\tilde{e}(\alpha) = \|e_V\|_2 \frac{e_V^\parallel + \alpha e_V^\perp}{\|e_V^\parallel + \alpha e_V^\perp\|_2}. \quad (7)$$

Decreasing α suppresses off-subspace error; the control exchanges $e_V^\parallel$ and $e_V^\perp$. Phase angles are held fixed and Mean PB is normalized at $\alpha = 1$. At $\alpha = 0$, off-subspace suppression lowers Relative Mean PB to 0.586, 0.476, 0.761, and 0.857 for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively, whereas the corresponding in-subspace controls raise it to 1.832, 1.316, 1.260, and 1.364.

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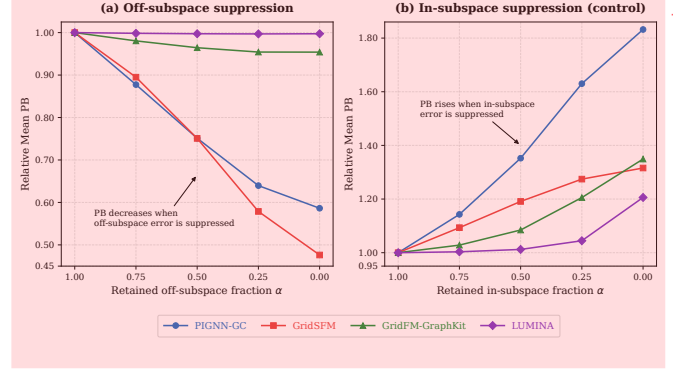


Fig. 2. Fixed-norm directional intervention on GBnetwork. Each curve is normalized by its $\alpha = 1$ Mean PB. Suppressing off-subspace error reduces PB, whereas suppressing in-subspace error produces the complementary increase.

Table 1. Basis controls on GBnetwork for PIGNN-GC (seed 42, $k = 16$). Random denotes the mean over eight independently sampled bases.

Projection	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$
None (C)	.01421	.753	.68522	.49329	.14511
Solution SVD	.01287	.822	.85299	.48903	.05690
Random	.02336	6.825	.00540	.00290	17.74
Laplacian LF	.02132	5.901	.20128	.12304	30.53
Spatial LP	.01568	1.926	.63080	.40717	29.29

PB to 0.586/0.476 for PIGNN-GC/GridSFM, whereas the control raises it to 1.832/1.316.

Consistent with the fixed-norm experiment, a complementary local Jacobian analysis shows that perturbations outside the training-solution subspace are far more strongly amplified into AC residuals than those inside it. Across 4,096 paired directions at 128 held-out points, median residual gain is 9.18 in-subspace versus 14,430 off-subspace (1,560 $\times$).

3.4. Specificity of the solution subspace.

To test whether the observed benefit follows from generic dimensionality reduction or graph smoothing, we replace the solution-SVD projection in CSP with matched random, graph-Laplacian, and spatial low-pass projections. As shown in Table 1, only the training-solution SVD improves both voltage-magnitude accuracy and power balance while increasing R_w^2 from .685 to .853 and nearly preserving a_w . In contrast, all three controls degrade scenario tracking and substantially increase AC residuals. Thus, the CSP gain cannot be explained by low dimensionality or spatial smoothing alone, but depends on the orientation of the training-solution subspace.

3.5. Post-training CSP correction

Table 2 compares raw predictions, calibration (C), projection (P_{16}), and CSP₁₆ on frozen model outputs. Relative to C , CSP reduces Mean PB by 67.0% for PIGNN-GC and 37.8% for GridSFM while lowering voltage-magnitude RMSE; angle RMSE increases slightly. Projection improves R_w^2 while largely preserving a_w . The lower tracking scores of gridfm-graphkit and LUMINA nevertheless show that reduced PB alone does not establish a useful scenario-dependent surrogate.

R29 At $k = 16$, the magnitude and angle subspaces retain 98.79% and 99.68% of held-out reference variation. In the seed-42 diagnostic, the removed magnitude-error energy exceeds the discarded reference variation by $6.46\times$ and $10.87\times$ for PIGNN-GC and GridSFM, respectively, explaining the magnitude improvement through Eq. (5). For angle, the inequality reverses, consistent with its slight degradation.

Applying magnitude and angle projection separately reveals model-dependent behavior. For PIGNN-GC, magnitude-only and angle-only projection yield Mean/Max PB of 0.0971/112.45 and 0.1499/218.12, respectively, both worse than full CSP. For GridSFM, magnitude-only projection gives the lowest PB among the projection variants (0.0458/2.90), whereas angle-only projection worsens PB to 0.0962/7.94.

R30 **Complementarity with one-step Newton refinement.** We next test whether geometric correction remains useful when followed by residual-directed refinement. Starting from frozen calibrated outputs (C) or CSP_{16} , we apply exactly one custom Newton update (NR_1), with validation-selected damping $\eta = 1$ fixed at test time. All pre- and post-update states are rescored using the same complex128 AC evaluator on 12,344 test scenarios per seed (41–43). No step failures or fallbacks occur.

R31 The benefit of CSP persists after Newton refinement (Table 3). For GridSFM, $C + NR_1$ reduces Mean PB from .08251 to .00250, yet the three-seed mean $|V|$ RMSE increases from .01086 to .01186, with an over-responsive slope ($a_w = 1.385$). Thus, residual reduction after one update need not improve reference-state accuracy. Starting from CSP_{16} instead yields $|V|$ RMSE .00097 and $a_w = .983$. Relative to $C + NR_1$, $CSP_{16} + NR_1$ reduces post-update Mean PB by 47.9% for PIGNN-GC and 51.2% for GridSFM, with lower magnitude and angle errors for both models. The corresponding p99 of scenario-wise maximum PB decreases from .774 to .499 and from .635 to .466, respectively.

R32 For 12,344 scenarios, CSP takes 2.03–2.07 s on an NVIDIA A40, whereas the custom Newton stage takes 2382–2391 s using 16 CPU workers. These implementation-specific correction times exclude neural inference and final metric scoring and are not hardware-normalized. Together, the results support CSP as a lightweight geometric correction that also improves initialization for one-step Newton refinement.

3.6. Cross-grid breadth and distribution shift

R34 **Cross-grid breadth.** Fig. 3 evaluates GridSFM with CSP across 31 systems spanning 4–9,241 buses and sub-kV to 750-kV levels: 28 pandapower/PYPOWER-format systems from IEEE, PEGASE, RTE, and GB benchmark families, plus three Common Information Model (CIM)-based networks. For each system, calibration and subspace statistics are fitted from its training data, using nominal $k = 16$ per channel and the available SVD components on smaller systems. CSP reduces GridSFM Mean PB across all 31 systems.

R35 **Topology and spatial shifts.** Under $N-1/N-2$ outages, tracking and bulk PB remain close to control, with larger maxima indicating isolated topology-induced violations and the need for worst-case assessment beyond average error [26]. GBcorr causes a broader shift: transferring both control calibration and basis gives Mean PB ≈ 0.100 p.u. Refitting calibration alone reduces this to 0.0944, basis alone to 0.0740, and both to ≈ 0.068 . The transferred basis retains 93.55% of GBcorr magnitude variation, versus 98.53% after refitting, while reference-state projection reduces Mean PB from 0.0868 to 0.0438 p.u. This shows that GBcorr introduces not only calibration drift but also solution-subspace mismatch.

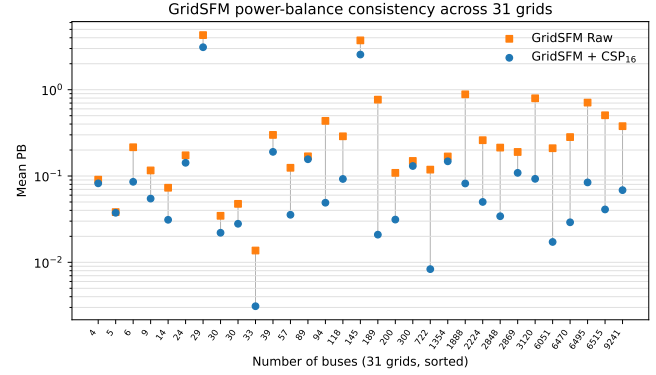


Fig. 3. Grid-by-grid physical consistency of GridSFM before and after CSP_{16} across 31 systems ordered by bus count. Mean PB is shown on a logarithmic scale.

4. CONCLUSION

This work identifies output-error geometry as an important factor in the accuracy–consistency gap of neural AC-PF surrogates. Fixed-norm and Jacobian analyses reveal strong directional sensitivity of AC residuals, while CSP reduces power-balance residuals and improves voltage-magnitude accuracy on GBnetwork. CSP also improves voltage accuracy and mean power-balance residuals after one Newton update, supporting its complementary role in residual-directed refinement. Cross-grid evaluation shows that the power-balance improvement extends beyond GBnetwork, whereas distribution-shift experiments show that a fixed solution subspace can become mismatched to the shifted operating distribution. These results motivate extending physics-consistent neural surrogates toward broader power-system analysis, estimation, and optimization.

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Table 2. GBnetwork results at $k = 16$ over three seeds (mean $\pm$ standard deviation). C denotes calibration, P_{16} projection, and CSP_{16} their composition. The per-bus mean baseline uses training-set means. Projection substantially improves R_w^2 while leaving a_w nearly unchanged, indicating that CSP removes unsupported directional components rather than recovering missing scenario-response amplitude.

Model	Output	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$
Per-bus mean	Baseline	.02341	6.847	.000	.000	.12084	5.88
PIGNN-GC	Raw	.02850 $\pm$.00232	1.108 $\pm$.175	.671 $\pm$.050	.546 $\pm$.072	.31592 $\pm$.03240	195.89 $\pm$ 52.54
	C	.01393 $\pm$.00105	.777 $\pm$.161	.671 $\pm$.050	.546 $\pm$.072	.16661 $\pm$.02352	190.00 $\pm$ 37.53
	P_{16}	.01624 $\pm$.00254	1.091 $\pm$.166	.829 $\pm$.045	.541 $\pm$.071	.07594 $\pm$.00422	5.07 $\pm$.85
	CSP_{16}	.01222 $\pm$.00134	.835 $\pm$.141	.829 $\pm$.045	.541 $\pm$.071	.05497 $\pm$.00257	3.93 $\pm$.07
GridSFM	Raw	.02868 $\pm$.01652	2.294 $\pm$.302	.768 $\pm$.151	.724 $\pm$.149	.26867 $\pm$.13004	9.10 $\pm$ 4.02
	C	.01086 $\pm$.00403	1.781 $\pm$.055	.768 $\pm$.151	.724 $\pm$.149	.08251 $\pm$.01093	7.29 $\pm$ 3.53
	P_{16}	.01814 $\pm$.01145	2.244 $\pm$.325	.936 $\pm$.032	.724 $\pm$.149	.07630 $\pm$.01942	5.08 $\pm$.49
	CSP_{16}	.00782 $\pm$.00307	1.807 $\pm$.053	.936 $\pm$.032	.724 $\pm$.149	.05136 $\pm$.00143	4.94 $\pm$.42
gridfm-graphkit	Raw	.00892 $\pm$.00154	.385 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.09282 $\pm$.01763	19.82 $\pm$ 8.65
	C	.00753 $\pm$.00118	.367 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.07685 $\pm$.01307	18.88 $\pm$ 8.69
	P_{16}	.00617 $\pm$.00099	.503 $\pm$.018	.959 $\pm$.008	.801 $\pm$.044	.04593 $\pm$.00156	5.36 $\pm$.56
	CSP_{16}	.00606 $\pm$.00090	.496 $\pm$.021	.959 $\pm$.008	.801 $\pm$.044	.04571 $\pm$.00135	5.08 $\pm$.18
LUMINA	Raw	.02689	2.347	.448	.358	.83072	464.63
	C	.01768	1.373	.448	.358	.22800	38.85
	P_{16}	.01943	1.856	.892	.350	.07713	4.83
	CSP_{16}	.01549	1.215	.892	.350	.06597	4.51

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Table 3. One Newton update from calibrated or CSP outputs on GBnetwork.

Model	Start	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$
PIGNN-GC	C	.00180 $\pm$.00012	.119 $\pm$.013	.975 $\pm$.001	.00284 $\pm$.00046	1.72 $\pm$.82
	CSP_{16}	.00117 $\pm$.00010	.073 $\pm$.009	.979 $\pm$.002	.00148 $\pm$.00013	.782 $\pm$.021
GridSFM	C	.01186 $\pm$.00677	.681 $\pm$.380	1.385 $\pm$.266	.00250 $\pm$.00154	.860 $\pm$.255
	CSP_{16}	.00097 $\pm$.00020	.059 $\pm$.013	.983 $\pm$.003	.00122 $\pm$.00029	.848 $\pm$.093

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Table 4. Fixed-rank CSP_{16} under topology and operating-distribution shifts. T transfers the control-derived calibration and solution basis, whereas R refits both on condition-matched training data. PB statistics are pooled over all bus-scenario pairs.

Condition	RMSE $_V$	R_w^2 $\uparrow$	a_w	Mean	Median	p95	p99	Max
Control	.01281	.8476	.4866	.0581	.0131	.2665	.5462	3.58
$N-1$ (T)	.01314	.8391	.4750	.0603	.0131	.2713	.5632	33.16
$N-2$ (T)	.01288	.8458	.4870	.0617	.0132	.2713	.5721	27.45
GBcorr (T)	.01966	.4819	.4025	.1003	.0231	.4512	.9746	7.00
GBcorr (R)	.01620	.5150	.4247	.0680	.0174	.3031	.6415	4.99

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Promising correction. Major revision required.

Physics-Informed but Not Physics-Consistent: Error Geometry and Subspace Projection for Neural AC Power Flow

Assessment. The core idea is useful: a cheap, grid-specific affine projection can suppress prediction components poorly supported by training solutions. Equation (5) is algebraically correct, and the Raw / C / P / CSP ablation is a good starting point. However, conflicting results, underspecified electrical conventions and incomplete controls currently limit the strength of the physical and practical conclusions.

What is established by this review. The original six pages were read and visually inspected. The algebra of Eq. (5), selected percentage calculations and a residual-cancellation counterexample were independently checked. Selected primary sources were consulted for PF conventions, dataset provenance, related work and submission requirements.

What is not established. No training, PF benchmark or repository implementation was reproduced. The manuscript’s code link returned 404 in the review-time retrieval. Comments marked as verification risks identify missing evidence; they are not allegations of leakage, incorrect code or fabricated results. This is a technical revision assessment, not a prediction of the conference decision.

How to use the annotated PDF

The first six pages preserve the manuscript and carry highlighted passages or outlined sections. Each margin label opens the matching entry in this index. The highlight itself contains a full PDF comment; an annotation-capable viewer can display it. All comment text is repeated here for viewers that do not show pop-ups.

CRITICAL	9	Submission blocker, internal contradiction or foundational check.
MAJOR	34	Important method, evidence, interpretation or reporting issue.
EDITORIAL	4	Writing, figure, citation or layout improvement.
VERIFIED	1	A checked result worth retaining.

48 indexed comments. The severity is the importance of resolving an issue; the status line separately identifies confirmed findings, proposed checks and interpretation. External citations [E1]–[E10] refer to the final source register.

Fix in this order

Priority	Minimum revision action	Comments
1 · Submission and consistency	Resolve page count and tiny tables; remove the duplicate subsection; regenerate Figure 2 and its endpoints; correct the gridfm-graphkit interpretation.	R01, R24, R25, R28, R43
2 · Electrical correctness	Specify free variables, residual masks, known-variable restoration and feature availability; demonstrate a tolerance-level residual on the unmodified reference.	R07, R19, R14, R38
3 · Evidence for geometry	Retain the verified state-error identity, but quantify residual cancellation, actual error directions and the reference variation destroyed by each projector.	R12, R15, R16, R26, R27, R29
4 · Reproducibility and scope	Fix dataset provenance; publish exact splits, seeds, losses and model modifications; distinguish target-grid fitting from zero-shot transfer.	R17, R18, R20, R40, R34, R35
5 · Practical usefulness	Add shifted Raw/C baselines and scenario-level tails; validate NR1; compare conventional starts and end-to-end timing.	R31, R32, R33, R44, R45
6 · Optional method improvement	Explore channel-wise ranks or shrinkage and a residual-based acceptance rule. These are proposed experiments, not results of the current paper.	R21, R30, R16

A compact regular-track structure

Page 1: problem, qualified contributions and a simplified overview. Page 2: electrical definitions, CSP and the state-error identity, followed by essential setup. Page 3: the main ablation and one reconciled geometry figure. Page 4: strongest Newton/shift evidence, limitations and conclusion. Page 5: references and permitted declarations. Compress the number of claims before compressing font size.

Several requested checks can use saved predictions and references rather than retraining: regenerate tables/figures, score x^* , audit masks, compute oracle-projection residuals, inspect cross terms and add scenario-level statistics. Broader pretraining and loss-weight comparisons are a second-stage extension.

Local editorial clean-up

Use “AC power-balance residuals” instead of “residuals from the AC power-balance”; “rank-k basis” instead of “SVD k rank basis”; and “reduced voltage error and residual” instead of “Reduced Small voltage Error and Residual.” Keep model names and hyphenation consistent, explain every abbreviation on first use, and use the same digits/seed labels in prose, figures and tables.

Comment index

Original manuscript · page 1

R01 | Regular-track submission format is not compliant

CRITICAL · Confirmed + external check · Manuscript p. 1

Finding. The supplied manuscript has six pages, and Tables 2–4 occupy page 5. For a regular ICASSP 2027 paper, the official paper kit permits four pages of technical content and one additional page restricted to references, funding acknowledgements and an ethics statement [E1]. This is a submission blocker, not a scientific objection. A separately selected OJSP route has different limits.

Revision. Confirm the submission track. For the regular track, remove the duplicated subsection, move all essential tables into pages 1–4 and compress the bibliography without shrinking the type. Do not submit this annotated review copy.

R02 | Distinguish adapted backbones from pretrained foundation models

MAJOR · Claim calibration · Manuscript p. 1

Finding. The abstract foregrounds foundation models, but Section 3.1 states that every model was trained from scratch and that GridSFM and LUMINA were adapted from AC-OPF. The experiment therefore evaluates specific AC-PF implementations, not the released pretrained systems. The original GridSFM description explicitly involves multi-grid pretraining [E7].

Revision. Say “four architectures trained from scratch for AC-PF, including adapted foundation-model backbones.” State which implementations actually use a physics loss, and avoid implying that the results establish a limitation of all pretrained or physics-informed solvers.

R03 | Separate the gain from calibration from the gain from projection

MAJOR · Interpretation · Manuscript p. 1

Finding. These are Raw-to-CSP comparisons. They correctly describe the combined method, but attribute neither part of the improvement specifically to subspace geometry. Table 2 already supplies the more informative calibrated baselines: PIGNN-GC 0.16661 → 0.05497 and GridSFM 0.08251 → 0.05136 p.u.

Revision. Report the incremental C-to-CSP reductions (67.0% and 37.8%) alongside, or instead of, the raw comparison. Specify voltage-magnitude accuracy rather than general voltage accuracy, because angle RMSE is not uniformly improved. Add the limitation that CSP reduces residuals without enforcing exact feasibility.

R04 | Simplify Figure 1 and qualify its causal labels

EDITORIAL · Figure interpretation · Manuscript p. 1

Finding. The figure is difficult to read at one-column size, and “Drives Physics Violation” presents the off-subspace direction as a universal cause. The schematic also visually suggests that projection simply removes the unknown error component. Equation (5) correctly shows that discarded reference variation also contributes to the final error.

Revision. Use a clear left-to-right workflow: raw prediction → bias correction → affine projection → residual evaluation. Mark the drawing as schematic; replace the causal label with “empirically higher residual sensitivity.” Show the reference tail or state that it is omitted for illustration. Enlarge labels rather than adding more elements.

Original manuscript · page 2

R05 | Make the novelty more precise than applying PCA after training

MAJOR · Positioning + external check · Manuscript p. 2

Finding. The distinction from Compact Learning is useful, but PCA compression and solver warm-starting are already present in that work [E6]. The strongest contribution here is the empirical link among error orientation, residual amplification and a frozen-model correction—not the existence of a low-rank voltage representation itself.

Revision. State three narrow contributions: a controlled error-orientation diagnostic; a bias-plus-affine-projection correction with an exact state-error trade-off; and evidence about grid-specific transfer and Newton initialization. Compare explicitly against PCA-output learning and residual-directed correction, without claiming a new feasibility projector.

R06 | Provide a reproducible, accessible release snapshot

MAJOR · Reproducibility / access limitation · Manuscript p. 2

Finding. The manuscript supplies a repository address, but the review-time retrieval returned HTTP 404. This may reflect a private, moved or not-yet-released repository; it does not establish that the code does not exist. The paper alone leaves several implementation-sensitive definitions unresolved.

Revision. Verify access from a logged-out browser and archive an immutable release. Include exact splits, sampled scenarios, model configurations, residual masks, basis files, seeds and scripts generating every figure/table from one results manifest. The present review did not reproduce training or audit repository code.

R07 | Write the AC-PF residual masks explicitly

CRITICAL · Definition requiring verification · Manuscript p. 2

Finding. The displayed residual stacks P and Q at every bus, whereas the conventional reduced polar PF problem uses active-power equations at non-slack PV/PQ buses and reactive-power equations at PQ buses. Slack injections and PV reactive injections are solved quantities, not independent prescribed setpoints [E3]. “Standard conventions” is insufficient to reproduce PB or the Jacobian.

Revision. Define scenario-specific masks M_P and M_Q , net-injection signs, slack treatment and $PV \leftrightarrow PQ$ switching. Specify whether a different full-state diagnostic is intentional. Unit-test the evaluator on the unmodified converged reference before interpreting residual gains. See the technical notes for a precise formulation.

R08 | Define units, aggregation and meaningful residual thresholds

MAJOR · Metric definition · Manuscript p. 2

Finding. It is unclear whether Mean PB averages per-bus two-component norms, components, per-scenario means, or all entries including masked zeros. Different bus-type mixes can then change the denominator, especially across 31 grids. The paper also does not specify the power base used for p.u. residuals.

Revision. Give PB_{is} , the evaluated bus set, Mean/Max formulas, per-grid baseMVA and units for all tables. Add the reference residual floor and scenario-level failure rates at prespecified tolerances. Convert to MVA only with the stated base. Distinguish these diagnostics from operational voltage, generator and branch-limit checks.

R09 | Explain the calibration estimator and its sampling bias

MAJOR · Reporting / statistical limitation · Manuscript p. 2

Finding. Training-only bias estimation avoids test-label leakage, which is a strength. However, residuals evaluated on examples used to fit the network can underestimate deployment bias. The text also does not say explicitly that this is one intercept per bus and channel, or whether circular angle offsets use a circular mean rather than an arithmetic mean of wrapped differences.

Revision. State the estimator, dimensions and angle procedure. Add a held-out calibration or out-of-fold sensitivity experiment without tuning on the test set. Keep the existing train-only result as the primary protocol if desired; do not call it leakage merely because the same training set supplies the basis.

R10 | Specify SVD orientation and effective rank

MAJOR · Notation / reproducibility · Manuscript p. 2

Finding. The proposed left-singular-vector projector is dimensionally correct only when buses are rows of X_V and X_{θ} . The matrix dimensions are not given. Also, k directions per channel produce total rank up to $2k$, not k . “Available components” is ambiguous when constant coordinates or redundant samples produce zero singular values.

Revision. Write $X_V \in \mathbb{R}^{(N \times n_{\text{train}})}$, $W_{V,k} \in \mathbb{R}^{(N \times k_V)}$, and analogous angle dimensions. Cap by numerical rank with a stated tolerance; report k_V and k_{θ} per grid. Distinguish the nominal label CSP16 from the effective retained rank.

R11 | Fix the angular chart and state-error scaling

MAJOR · Coordinate definition · Manuscript p. 2

Finding. Gauge alignment removes an arbitrary reference offset but does not, by itself, resolve phase wrapping. A Euclidean SVD across a $\pm\pi$ discontinuity and the Euclidean identity in Eq. (5) need a consistent unwrapped chart. A concatenated magnitude/angle norm also mixes p.u. and angular units.

Revision. State radians inside trigonometric functions/Jacobians and degrees only for reporting, identify the reference bus per island, and describe unwrapping. Analyse Eq. (5) separately by channel or declare a fixed scaling metric. An orthogonal projector in scaled coordinates must be interpreted in that same metric.

R12 | A solution covariance subspace is not a fixed-input feasible tangent

MAJOR · Mechanistic inference · Manuscript p. 2

Finding. The SVD spans variation across different operating conditions. Such directions are not generally null directions of the fixed- u residual Jacobian. Differentiating $r(x^*(u), u) = 0$ gives $J_x dx^* + J_u du = 0$. Dominant state covariance can therefore reflect the input sampling covariance as well as network sensitivity.

Revision. Keep the explanation empirical rather than identifying the SVD subspace with the physical tangent or a universal low-gain space. Vary the injection covariance and compare singular spectra/principal angles. The technical notes show how the input covariance enters the linearized state covariance.

R13 | State the conditions behind the Taylor expansion

MAJOR · Assumptions / evaluator check · Manuscript p. 2

Finding. Equation (3) is the standard local expansion and its sign is consistent with defining J as the derivative of r . However, a numerical NR reference has a tolerance-level residual rather than an exact zero, and the expansion assumes a fixed smooth residual definition. Bus-type switching, islanding or voltage-dependent injections require care.

Revision. Report the measured reference residual floor in the identical evaluator and the solver settings [E4]. State the active bus masks and operating regime for each Jacobian. Check analytic J_v against finite differences and show when the first-order approximation is accurate at the tested perturbation scale.

R14 | Preserve known voltages and the slack angle under CSP

CRITICAL · Potential implementation failure · Manuscript p. 2

Finding. An unconstrained affine projection of $[V, \theta]$ need not retain the scenario-specific PV voltage setpoints, slack magnitude or reference angle. This is especially important because the data vary PV setpoints. If PB omits PV reactive equations, low PB can coexist with a wrong PV voltage unless known variables are enforced separately.

Revision. Specify restoration/masking after both calibration and projection for every model and control. Prefer a projection in free PF coordinates, with known variables reconstructed from u . Re-derive the relevant error identity if post-projection masking changes the operator. Check boundary violations explicitly rather than assuming the training subspace preserves them.

R15 | Equation (5) is correct; clarify precisely what it proves

VERIFIED · Algebra checked · Manuscript p. 2

Finding. Let $Q = I - P$, $d = x^* - \mu$ and $e = e_C$. Then $e_{CSP} = P e - Q d$. Orthogonality gives $\|e_{CSP}\|^2 - \|e\|^2 = \|Qd\|^2 - \|Qe\|^2$, exactly as claimed. This is a valid and useful pointwise identity for the stated Euclidean affine projection.

Revision. Keep the identity and add this one-line derivation. Say “squared state error” rather than leaving “accuracy” unqualified. The same result can be averaged or applied by channel, but it does not imply that nonlinear Mean PB or Max PB decreases. Do not present the verified identity as a physics-feasibility theorem.

R16 | Account for residual-space cancellation, not only directional gain

MAJOR · Mathematical limitation · Manuscript p. 2

Finding. Even when the reference lies in the retained affine subspace, $\|JPe\|^2 - \|Je\|^2 = -\|JQe\|^2 - 2(JPe, JQe)$. A negative cross term can make projection increase residuals while reducing state error. Orthogonality of Pe and Qe does not imply orthogonality after multiplication by J .

Revision. Retain the existing no-guarantee statement and make this mechanism explicit. Measure the residual cross term on actual model errors, not only random probe gains. The technical notes include an invertible-Jacobian counterexample and an optional residual-based acceptance rule.

R17 | Clarify the provenance of the 2,224-bus dataset

MAJOR · Citation-support mismatch · Manuscript p. 2

Finding. This sentence attaches reference [22] to the GBnetwork experiment. The authors' PFDelta repository lists IEEE 14/30/57/118 and GOC 500/2000 cases, not a 2,224-bus GBnetwork [E5]. That does not rule out an author-generated extension, but the citation alone does not establish the stated dataset provenance.

Revision. State whether GBnetwork scenarios are newly generated, adapted from PFDelta sampling code, or obtained from another source. Cite the actual network and generation procedure, and release the scenario manifest. Distinguish borrowing a benchmark methodology from using its released cases.

Original manuscript · page 3

R18 | Document the model modifications and physics-training objectives

MAJOR · Reproducibility · Manuscript p. 3

Finding. The new PIGNN-GC extension and phrases such as “stable initialization” and “enhanced AC-PF configuration” are not defined well enough to reproduce the comparison. Architectures and names do not establish that the same variables, loss masks or physics weights are used.

Revision. Provide the GC equations, attention/message-passing details, parameter counts, selected toolkit model class, inputs/outputs, loss formulas and normalization, lambda values/schedules, optimizer, training budget and checkpoint selection. Show supervised-only versus physics-penalized training or a small physics-weight sweep to directly support the title's physics-informed premise.

R19 | Audit which PPC values are available before solving PF

CRITICAL · Risk to verify; not an allegation · Manuscript p. 3

Finding. Using one internal representation improves consistency, but a solved PPC can contain quantities unavailable to a surrogate before PF, including solved slack dispatch and PV reactive generation. The manuscript does not provide a feature-by-bus-type availability table, so the absence of answer leakage cannot be checked from the text.

Revision. List each feature and whether it is a prescribed input, a derived topology quantity or a PF output. Construct inputs before the reference solve and assert that no solved unknown enters the model. Keep residual-scoring setpoints and known-variable masks separate from supervision targets.

R20 | Give exact split sizes and scenario-generation coverage

MAJOR · Reproducibility / scope · Manuscript p. 3

Finding. The approximate 1:1:1 split of 37,022 cases is not reconciled with 12,344 test cases per seed later in the paper. Filtering or distinct manifests could explain this; it is not evidence of leakage. Ranges, correlations and rejection rates for loads, generation and PV voltages are also absent.

Revision. Report exact train/validation/test counts and split seeds, common splits across models, sampling distributions, attempted versus converged cases, deduplication/grouping and reference solver tolerances. State explicitly that conclusions concern the sampled converged operating distribution, not all grid states or collapse-boundary cases.

R21 | Make the rank-selection rule auditable

MAJOR · Selection protocol · Manuscript p. 3

Finding. It is unclear which model or aggregation selected $k=16$ and whether “without degrading calibrated voltage RMSE” refers only to magnitude. A common rank may be a controlled design choice, but it is not established as optimal across models, both channels and all grid sizes.

Revision. Publish the validation curves and selection constraint, including the baseline used to judge degradation. Add $k=0$ and a no-truncation endpoint where meaningful, and consider separate k_V/k_θ or shrinkage chosen on validation only. Freeze every choice before test and shift evaluation, as the manuscript already intends.

R22 | Squared correlation and response amplitude need clearer interpretation

MAJOR · Metric correctness · Manuscript p. 3

Finding. R_w^2 is squared Pearson correlation, not the usual coefficient of determination. It discards sign; anti-correlated predictions can score one. The stated a_w is a regression slope $\text{Cov}(\text{true}, \text{pred})/\text{Var}(\text{true})$, not the general variance-amplitude ratio. For a per-bus constant predictor, correlation is undefined, although Table 2 records zero.

Revision. Rename the metric ρ_w^2 or explicitly distinguish it from regression R^2 ; report signed correlation and state the zero-variance convention. Define which samples supply the centering means and how buses are pooled. Add centered prediction R^2 or $\sigma_{\text{pred}}/\sigma_{\text{true}}$ if scenario amplitude is a key claim; a_w has target 1, not “larger is always better.”

R23 | Specify the oracle intervention and its physical admissibility

MAJOR · Diagnostic definition · Manuscript p. 3

Finding. Equation (7) uses x^* to construct a counterfactual error and is therefore an oracle diagnostic, not a deployable correction. That is appropriate if stated. It is unclear whether the unchanged angles are calibrated predictions or reference angles, whether normalization is per scenario, and how zero denominators are handled. Moving voltage errors can also alter prescribed PV magnitudes.

Revision. Define the evaluated state as $[V^* + e_{\text{tilde}}(\alpha), \theta_C]$ or the intended alternative. Restrict perturbations to free coordinates, specify epsilon/degenerate cases, and add joint-angle and reference-angle controls plus perturbation-scale checks. Describe α as a pre-renormalization coefficient, not the final retained energy fraction.

R24 | Remove the duplicated subsection and equation

CRITICAL · Confirmed editorial error · Manuscript p. 3

Finding. Sections 3.2 and 3.3 repeat the same heading, setup and normalized error construction; Equations (7) and (8) are duplicates. The second subsection also restates only part of the results, making the narrative look like an unresolved merge of manuscript versions.

Revision. Keep one directional-intervention subsection and one equation. Move the unique Jacobian paragraph into a clearly named follow-on paragraph or subsection. Regenerate numbering and references automatically. Reclaim this space for the missing residual definition and the main results table.

R25 | Figure 2 disagrees with the reported endpoints

CRITICAL · Confirmed figure/text inconsistency · Manuscript p. 3

Finding. At $\alpha=0$, panel (a) visually places gridfm-graphkit near 0.95 and LUMINA near 1.00, not the text's 0.761 and 0.857. Panel (b) is near 1.35 and 1.21 for these models, rather than 1.260 and 1.364. These are approximate readings from the rendered plot; the PIGNN-GC and GridSFM endpoints appear consistent.

Revision. Reconcile the figure, prose, seeds and evaluation cohort using the underlying results. Do not select one version by visual guess. Export plot data and generate reported endpoints from the same manifest. This discrepancy affects the claim that the directional effect is substantial across all four models.

R26 | The 1,560× gain claim needs an exact, reproducible definition

MAJOR · Evidence requiring validation · Manuscript p. 3

Finding. The text omits the definition and units of gain, whether directions perturb magnitude only or the concatenated state, how directions are sampled, and how 4,096 pairs are aggregated over 128 points. Very large gains can reflect scaling, near-zero impedances, disallowed boundary-coordinate perturbations or stiff network modes; they are not automatically an error.

Revision. Define $g(d) = ||jd||/||d||$ with its precise norms and masks. State the pair construction and whether 1,560× is the median paired ratio or a ratio of medians. Report quantiles, finite-difference checks, gain on actual model errors and a low-impedance/bus-type diagnostic. Include the residual-space cross term.

R27 | Rank-matched controls do not isolate physics-specific geometry

MAJOR · Control-design limitation · Manuscript p. 3

Finding. A random rank-16 basis need not retain the reference signal; PCA is designed to do so. The controls' angle errors approach the per-bus-mean baseline and PB rises to 17–31 p.u. This demonstrates poor replacement projectors, but does not by itself distinguish a physics-specific mechanism from better data reconstruction.

Revision. For every basis, report retained reference energy and PB after projecting x^* . Match reconstruction quality as a second comparison, preserve known variables, and define electrical Laplacian weights/spatial coordinates and filtering. Add a simple shrinkage or PCA-coefficient regression baseline. Narrow “cannot be explained” to what these particular controls exclude.

R28 | The gridfm-graphkit tracking narrative contradicts Table 2

CRITICAL · Confirmed table/text inconsistency · Manuscript p. 3

Finding. The final sentence groups gridfm-graphkit with LUMINA as having lower tracking scores. Yet Table 2 gives gridfm-graphkit the highest CSP squared correlation (0.959) and the response slope closest to one (0.801) among the four models. LUMINA's slope of 0.350 supports the amplitude caveat; the stated grouping does not.

Revision. Replace the sentence with a model-specific interpretation: CSP improves correlation without restoring missing response amplitude, most clearly for LUMINA. Acknowledge gridfm-graphkit's strong relative tracking. Also compare calibrated baselines, rather than presenting a model-quality ranking from inconsistent prose.

Original manuscript · page 4

R29 | High retained state variance is not a residual guarantee

MAJOR · Mechanism / reporting · Manuscript p. 4

Finding. The retained-variance and $6.46\times/10.87\times$ diagnostics are helpful, but their denominators, aggregation and cohort are not defined. A tiny discarded reference tail can still be amplified strongly by J. Also, the later 98.53% retained magnitude variation differs from 98.79%; different data/seeds could explain it.

Revision. State whether ratios are ratios of summed energies or means of per-scenario ratios, with the seed and split. Show distributions of off-subspace model-error energy, discarded reference energy and oracle-projection PB. Reconcile the retained-variance figures by naming their cohorts; do not infer residual safety from variance explained alone.

R30 | Use the channel ablation to reconsider the default correction

MAJOR · Ablation interpretation · Manuscript p. 4

Finding. GridSFM's magnitude-only correction has lower reported Mean/Max PB (0.0458/2.90) than full CSP, and angle-only projection worsens PB. Thus equal-rank projection of both channels is not uniformly supported by the objective used to select the method. Separate SVDs also discard magnitude-angle cross-covariance.

Revision. Compare calibrated, magnitude-only, angle-only and joint CSP with all state metrics on the same seeds. Select channel ranks or shrinkage on validation; never pick the test-best ablation retrospectively. A scaled joint basis is a useful optional control, but must respect units and known variables.

R31 | One custom Newton step is not a complete solver benchmark

MAJOR · Practical evaluation · Manuscript p. 4

Finding. This is a useful initialization experiment, not a comparison with a converged conventional solver. “No step failures” only means that a step could be executed; it does not imply tolerance-level convergence or operational feasibility. Mean PB and Max PB remain nonzero after the step.

Revision. Include flat/DC initialization, the static per-bus mean, C and CSP under the same sparse Newton implementation. Report one-step outcomes and time/iterations to a fixed residual tolerance, including failures. State variable ordering, Jacobian construction, bus-limit handling and the validation set used to choose damping.

R32 | Explain residual improvement with worsening GridSFM state error

MAJOR · Diagnostic requiring verification · Manuscript p. 4

Finding. Table 3 reports lower PB but worse magnitude RMSE for C+NR1 than C, including slope 1.385. This can occur for an incomplete Newton step, ill-conditioning or different feasible roots, but the present evidence does not distinguish these from masking, scaling or implementation issues.

Revision. Validate the custom update against a trusted PF implementation from exactly the same initial state. Show seed-level and scenario-level trajectories, the full masked residual, boundary errors and final converged roots. Avoid explaining the anomaly as an intrinsic physical phenomenon until these checks agree.

R33 | Benchmark the full pipeline on comparable hardware

MAJOR · Performance interpretation · Manuscript p. 4

Finding. The paper appropriately acknowledges the A40-versus-CPU mismatch and excluded stages. These batch totals nevertheless do not establish an algorithmic speedup over an optimized PF solver. Custom multiprocessing, data transfer and evaluator overhead can dominate the Newton timing.

Revision. Report CPU-CSP and a compiled/sparse NR baseline on the same machine, plus end-to-end neural inference, correction, transfers and residual checking. Separate batch throughput from single-scenario latency and include basis fitting/storage costs. Apply $W(W^T z)$ in $O(Nk)$ rather than materializing a dense $O(N^2)$ projector.

R34 | Distinguish breadth across grids from unseen-grid transfer

MAJOR · Generalization scope · Manuscript p. 4

Finding. Every grid receives training-derived calibration and basis statistics. This establishes grid-wise applicability with target-grid reference data, not a frozen, zero-shot CSP operator for unseen grids. It is also unclear whether the neural backbone is trained separately per grid or shared across all 31.

Revision. State the training/transfer protocol for the backbone and correction independently, list grids and per-grid sample counts, and report target-grid labeling cost. Use “grid-wise evaluation with per-grid fitting” unless a distinct held-out-grid experiment is supplied.

R35 | On small grids the nominal projection may be the identity

MAJOR · Attribution confound · Manuscript p. 4

Finding. For systems with $N \leq 16$, retaining every available bus-space component can make $P=I$ (or identity on all admissible variable directions). Then CSP reduces to calibration, so a Raw-to-CSP improvement is not evidence of projection benefit. Figure 3 includes 4-, 5-, 6-, 9- and 14-bus systems.

Revision. Plot Raw, C, P and CSP for each grid, and show effective ranks with the numerical-rank convention. Separate the bias-only small-grid regime from genuine truncation. Report paired relative improvements and uncertainty rather than inferring that projection itself improves all 31 systems.

R36 | Explain bus correspondence and topology-specific bases

MAJOR · GNN deployment limitation · Manuscript p. 4

Finding. The dense bus-indexed mean, bias and basis are tied to a specific coordinate ordering and node set. A GNN backbone’s permutation equivariance does not automatically make a saved CSP artifact portable across bus renumbering, bus fusion, islands or networks of a different size.

Revision. Persist the physical-bus-to-PPC mapping with every artifact. Under a bus permutation, transform the mean, bias and basis rows consistently, and test that the corrected output permutes accordingly. Explain what is rebuilt when the node set or slack reference changes. This is a deployment limitation, not a claim that CSP is inherently non-equivariant.

R37 | Define GBcorr and the outage protocols

MAJOR · Missing experimental definition · Manuscript p. 4

Finding. GBcorr is not operationally defined in the paper. The correlation structure, perturbation scale, changed quantities and number of cases are unknown. The N–1/N–2 protocols also omit branch sampling, islanding treatment, convergence filtering and whether the admittance and bus masks are updated per outage.

Revision. Provide a compact protocol table for each shift, with sampling formulas and accepted/rejected counts. Specify whether the frozen network receives the new topology and whether R refits only CSP statistics. Separate zero-shot transfer from adaptation using condition-matched solved labels, with a labeling-budget curve if feasible.

R38 | Disambiguate the nonzero “reference-state projection” residual

CRITICAL · Ambiguity; not a proven evaluator bug · Manuscript p. 4

Finding. This wording can be read as saying that the converged reference itself has Mean PB 0.0868, conflicting with $r(x^*;u) \approx 0$. A plausible intended meaning is that projecting the reference with the transferred basis gives 0.0868, whereas the refitted basis gives 0.0438. These are very different claims.

Revision. Write all three states explicitly: x^* , $\mu_{\text{control}} + P_{\text{control}}(x^* - \mu_{\text{control}})$, and $\mu_{\text{shift}} + P_{\text{shift}}(x^* - \mu_{\text{shift}})$. Report PB for each in the same evaluator. The unprojected reference should match the configured solver/evaluator tolerance. Resolve any nontrivial reference residual before drawing geometric conclusions.

R39 | Keep the conclusion within the demonstrated application scope

EDITORIAL · Claim calibration · Manuscript p. 4

Finding. The results support a lightweight, data-supported correction and an initialization aid, not an exactly feasible solver. The last sentence reaches toward estimation and optimization without experiments in those tasks. The manuscript does acknowledge its main limitation in Section 2.3; preserve that distinction consistently.

Revision. End with the conditions under which CSP helps: representative target-grid reference data, compatible coordinates, acceptable discarded signal and favorable residual sensitivity. Present estimation/optimization as future work. Mention remaining tail violations and model-dependent angle trade-offs instead of calling corrected surrogates physics-consistent without qualification.

Original manuscript · page 5

R40 | The three-seed caption does not explain LUMINA’s uncertainty

MAJOR · Confirmed reporting inconsistency · Manuscript p. 5

Finding. The caption says mean $\pm$ standard deviation over three seeds, but LUMINA has only single values. The PDF does not establish whether these are single-seed results or means with omitted deviations. Some other diagnostics use seed 42 only, whereas Tables 2–3 summarize several seeds.

Revision. Label the exact seed count for every row and every table. Add LUMINA deviations or identify the one-seed exception. Show paired seed-wise improvements and scenario-resampled intervals; bus-scenario entries within a grid are not independent replicates. Do not equate millions of bus evaluations with millions of independent experiments.

R41 | Residual reduction is substantial but is not a feasibility certificate

MAJOR · Operational significance · Manuscript p. 5

Finding. The CSP rows still have mean PB around 0.046–0.066 p.u. and maximum PB around 3.9–5.1 p.u. The per-bus-mean baseline also has a moderate mean residual despite no scenario response. These reported values show why improvement percentages and small voltage RMSE alone do not establish operational usefulness.

Revision. Declare application-relevant tolerances and report the fraction of scenarios satisfying all tested conditions, not just average PB. Add branch-flow/current accuracy and limit violations if operational claims are retained. The right interpretation is “lower residuals,” not “AC-feasible outputs.”

R42 | The angle trade-off is material and differs by model

MAJOR · Quantitative interpretation · Manuscript p. 5

Finding. For gridfm-graphkit, C-to-CSP angle RMSE increases from 0.367° to 0.496°, about 35.1%, even though magnitude RMSE and PB improve. PIGNN-GC and GridSFM also worsen in angle, while LUMINA improves. Thus the trade-off is not uniformly a “slight” increase across all models.

Revision. State absolute and relative changes by model, with uncertainty. Avoid saying all voltage errors improve. Connect this result to the channel-wise Eq. (5) diagnostic and magnitude-only correction, rather than burying it in a generic caveat.

R43 | Table 3 is far too small to read reliably

CRITICAL · Confirmed typography + external check · Manuscript p. 5

Finding. The embedded body text in Table 3 is approximately 5.26 pt (smaller superscripts are about 3.94 pt), while the official paper kit asks for at least 9 pt [E1]. It is visually illegible at normal page size. Other reduced table/figure labels should also be checked.

Revision. Rebuild Table 3 at a readable size, preferably spanning both columns or sharing a compact table with Table 2. Reduce redundant columns/digits instead of globally scaling a tabular environment. Resolve page count by editing content, not by shrinking fonts.

R44 | Identify the Table 4 model and add shifted baselines

MAJOR · Incomplete reporting / comparison · Manuscript p. 5

Finding. The caption does not identify the neural model, seed count, scenario count or meaning of the unlabelled PB units. All listed rows are CSP variants, so the table cannot show whether CSP helps relative to Raw or C under each shift. T versus R also compares different access to target-distribution reference labels.

Revision. Name the model and cohort and cite Table 4 explicitly in the shift discussion. Include Raw, C and CSP on the same shifted test scenarios, plus calibration-only and basis-only refits where claimed. State the adaptation-data budget and keep it disjoint from test data.

R45 | Pooled bus quantiles conceal scenario-level failures

MAJOR · Reliability assessment · Manuscript p. 5

Finding. Table 4 pools all bus-scenario pairs. Its N–1 p99 PB is 0.5632 while the maximum is 33.16, versus control max 3.58. These numbers are compatible with a small subset of severe failures, but pooled quantiles do not tell how many scenarios are affected or where violations occur.

Revision. Add quantiles of each scenario’s maximum PB, the proportion exceeding a fixed tolerance, and the top offending buses/branches. Use common outage cohorts when comparing N–1 and N–2 maxima. Avoid reading similar pooled means as contingency robustness.

R46 | The package checksum overflows the reference column

EDITORIAL · Confirmed layout issue · Manuscript p. 5

Finding. Reference [15] prints a long unbreakable SHA-256 string to roughly $x=604$ pt on a 612-pt-wide page, well beyond the ordinary right text margin. It consumes space while still failing to identify the actual architecture/class used from the toolkit.

Revision. Move the checksum to the reproducibility manifest and retain a compact, versioned software citation with an accessible source identifier. State the selected model class and configuration in the methods. Audit reference wrapping, reduce redundant author-list space using the venue style, and regenerate figures without tiny or Type 3 text where possible.

Original manuscript · page 6

R47 | Keep bibliography validation separate from citation relevance

EDITORIAL · Limited external spot-check · Manuscript p. 6

Finding. Selected primary records support the existence of Compact Learning, GridSFM, LUMINA, GENCO and the cited autoencoder-projection work [E6–E10]. This review did not verify every bibliography field or every citation. The concrete issue with [22] is support for the particular GBnetwork dataset, not evidence that PFDelta is fabricated.

Revision. Check each remaining identifier, author list, venue/year and the exact claim it supports against the primary record. Explain relevant differences from both linear PCA reconstruction and nonlinear feasibility corrections. Do not replace valid preprint citations solely because they are preprints.

R48 | Add ethics and conflict-of-interest declarations

MAJOR · External submission-policy check · Manuscript p. 6

Finding. The supplied PDF has a funding acknowledgement but no explicit ethics-compliance statement or statement about conflicts of interest. The ICASSP 2027 policy asks for both, including when ethical approval was unnecessary [E2]. The correct statements depend on the authors' actual circumstances.

Revision. Add a short "Compliance with Ethical Standards" section and an accurate conflict-of-interest disclosure. A numerical-simulation/no-human-or-animal-subject statement is suitable only if true for all data used. The permitted fifth-page material includes these declarations under the paper kit; verify the chosen submission route.

Technical notes: what the geometry does prove

The following derivations are reviewer analysis of the stated model. They are not additional experiments, empirical certificates or claims made by the authors.

1. Use an explicitly reduced PF formulation

For a conventional connected grid with fixed bus types, the free coordinates may be written as $z=[\theta \text{ at PV/PQ buses}; V \text{ at PQ buses}]$. Known PV/slack magnitudes and the reference angle belong to u and are reconstructed separately. The corresponding residual is $[P_{\text{set}}-P \text{ at PV/PQ}; Q_{\text{set}}-Q \text{ at PQ}]$. A different formulation is possible, but must be stated consistently [E3].

For a bus-wise presentation, define binary masks $m_{P,i}$ and $m_{Q,i}$ and score $PB_i = \sqrt{(m_{P,i} \Delta P_i)^2 + (m_{Q,i} \Delta Q_i)^2}$. Specify the set over which it is averaged; excluding slack-only zero entries avoids a hidden denominator change. Also assess any known-variable violations. Masked PB alone does not check a PV voltage setpoint or an operating limit.

Validate $r(x^*;u)$ with the exact evaluation model, precision, power base, injection convention and solver tolerance used for predictions. State voltage-dependent-load and reactive-limit options rather than relying on software defaults [E4]. If this baseline fails, geometric interpretations should wait until the discrepancy is resolved.

2. The affine-projection identity is exact

Let $P=P_k$ be orthogonal, $Q=I-P$, $\mu=x_{\text{bar_train}}$, $d=x^*-\mu$ and $e=x_C-x^*$. The corrected prediction is $x_{\text{CSP}}=\mu+P(x_C-\mu)$. Direct substitution gives:

$$\begin{aligned} e_{\text{CSP}} &= P e - Q d \\ \|e_{\text{CSP}}\|^2 - \|e\|^2 &= \|Q d\|^2 - \|Q e\|^2 \end{aligned}$$

The cross term vanishes because $\text{range}(P)$ is orthogonal to $\text{range}(Q)$. Equation (5) is therefore correct. The state-error improvement condition is exactly that discarded error energy exceeds discarded reference energy, in the same coordinate norm. Averaging the identity is valid; substituting an arbitrary weighted norm or applying additional nonlinear corrections requires a new derivation.

3. Why smaller state error can have a larger residual

In a fixed smooth regime, $r(x^*+e;u)=r(x^*;u)+J e+O(\|e\|^2)$. Even if $Qd=0$ and the reference residual is negligible, the linearized change in squared residual is:

$$\|JP e\|^2 - \|J e\|^2 = -\|JQ e\|^2 - 2(JP e, JQ e)$$

The last term can defeat the first. A nonsingular illustrative Jacobian is $J=[1,2],[0,0.1]$, with $P=\text{diag}(1,0)$, $d=0$ and $e=t(2,-1)$. Projection lowers squared state error from $5t^2$ to $4t^2$, while the residual norm increases from $0.1|t|$ to $2|t|$. The off-subspace unit direction even has greater gain than the in-subspace direction. This is a generic local counterexample, not a constructed AC-grid instance.

With a nonzero reference tail $q=Qd$, the linearized residual difference becomes $\|Jq\|^2 - \|JQe\|^2 - 2(JPe, J(q+Qe))$. Neither this Euclidean residual calculation nor Eq. (5) automatically establishes a decrease in the paper's mean of per-bus norms or its maximum.

Technical notes: strengthen the causal evidence

4. Separate network sensitivity from input covariance

Assume a local smooth solution map with an invertible reduced J_x . Implicit differentiation and first-order covariance propagation give:

$$K = -J_x^{-1} J_u$$

$$\text{Cov}(x^*) \approx K \text{Cov}(u) K^T$$

Thus the dominant SVD directions depend jointly on the physical map and the sampling distribution. In the special case of isotropic injection perturbations with $J_u=I$, large state-variance directions align with small singular-value directions of J_x . This gives a plausible local explanation for the empirical observation. With anisotropic or correlated injections, that relationship can change; it is not a universal property of solution PCA.

A decisive diagnostic package

For each model and seed, report the off-subspace fraction $\|Q_e\|^2/\|e\|^2$, the discarded reference energy $\|Q_d\|^2$, the gains on the actual P_e and Q_e , and the cosine between J_{P_e} and J_{Q_e} . Summarize per-scenario distributions rather than only ratios of aggregate energies. Show the residual of projected references for SVD and each control, with identical masks and boundary handling.

Pair this with a perturbation-scale sweep comparing exact nonlinear residuals with J_e . Repeat on genuinely free coordinates and inspect which branches/buses dominate the high-gain directions. Then change the injection covariance while keeping the network fixed. These tests distinguish state compression, stiffness, coordinate artifacts and residual cancellation without assuming which explanation wins.

5. Two optional, testable improvements

Channel-wise soft projection. Instead of deleting the tail, use $x_{\text{beta}} = \mu + [P + \beta Q](x_C - \mu)$, with $0 \leq \beta \leq 1$ and separate validation-selected values for magnitude and angle. Here $\beta=0$ is CSP and $\beta=1$ is C; the error is $P_e + \beta Q_e - (1 - \beta)Q_d$. This exposes a controllable bias-variance trade-off and is directly motivated by the reported angle degradation. Apply it in a boundary-compatible coordinate system. No performance improvement is claimed without testing.

Residual-based acceptance. After known-variable enforcement, evaluate C and CSP on the same scenario. Accept CSP only if both the selected mean PB and maximum PB do not exceed C, and any required hard-variable checks pass; otherwise retain C. This construction gives a pointwise non-worsening rule for those measured residual summaries relative to C. It does not guarantee smaller state error, exact feasibility or satisfaction of untested limits, and the two residual evaluations must be included in runtime.

6. Minimum practical comparison

Evaluate the same residual tolerance and hardware for flat/DC start, per-bus-mean start, C and CSP. Report one-step results and convergence-to-tolerance results, including unsuccessful cases. Under each shift, compare Raw, C and CSP before considering any refit, then label calibration/basis refits as adaptation with target-distribution labels. Keep scenario-wise maxima and failure rates alongside pooled means.

Suggested writing replacements

These drafts use only the current manuscript's reported results and avoid the disputed Figure 2 endpoints. Update all numbers after the evaluator and results manifest have been checked.

A more tightly scoped title

Error Geometry and Calibrated Subspace Projection for Neural AC Power Flow

The existing contrastive title can also work, provided the paper identifies which trained implementations were physics-informed and avoids generalizing to all such methods.

Revised abstract

Neural AC power-flow surrogates can predict voltages accurately while retaining substantial power-balance residuals. We investigate this gap using four architectures trained from scratch for voltage-state prediction on a 2,224-bus GBnetwork dataset. A training-derived solution subspace provides coordinates for analysing error orientation. We introduce calibrated solution-subspace projection (CSP), a non-iterative correction combining train-estimated bias removal with affine projection. Relative to bias calibration alone, CSP reduces mean power-balance residuals from 0.16661 to 0.05497 p.u. for PIGNN-GC and from 0.08251 to 0.05136 p.u. for GridSFM, while improving voltage-magnitude RMSE. Additional experiments examine per-grid refitting across 31 systems, distribution shifts and one-step Newton refinement. CSP does not enforce AC feasibility; its benefit depends on discarded reference variation and the sensitivity of AC residuals to the removed prediction components.

Replace the tracking interpretation

"Projection improves the correlation of scenario-dependent voltage variations while leaving their response slopes nearly unchanged. In particular, LUMINA retains a slope of 0.350 after CSP, showing that higher correlation and lower PB do not necessarily restore the amplitude of the scenario response. gridfm-graphkit has the strongest CSP tracking scores among the four evaluated implementations."

Replace the geometry-to-physics transition

"The training-solution basis defines data-supported coordinates rather than the feasible set for a fixed operating condition. Equation (5) characterizes the change in squared state error exactly. Whether this correction reduces power-balance residuals additionally depends on the residual Jacobian, discarded reference variation, cancellation between residual components and nonlinear effects. We therefore assess residual improvement empirically."

Replace the cross-grid claim

"We evaluate grid-wise applicability on 31 systems, fitting the calibration and solution basis separately from each system's training references. This experiment measures the breadth of the correction with target-grid reference data; it does not establish zero-shot transfer of a fixed correction operator to an unseen grid."

External source register and review boundaries

Primary records consulted on 13 September 2026. These checks supplement the supplied manuscript; they do not replace missing experimental evidence. References below are clickable. All manuscript-specific numbers and inconsistencies originate in the six-page attachment, at the locations identified by the indexed comments.

[\[E1\] ICASSP 2027 official paper kit](#)

Regular-track length, permitted fifth-page material and minimum type size. The OJSP route is separate.

[\[E2\] ICASSP 2027 conference policies](#)

Ethical-compliance and conflict-of-interest reporting requirements.

[\[E3\] MATPOWER manual, Section 4.1: AC Power Flow](#)

Conventional reduced polar variables, PV/PQ/slack equations and known quantities.

[\[E4\] pandapower documentation: Balanced AC Power Flow](#)

PPC access, solver options, mismatch tolerance and reactive-limit/voltage-dependent-load settings; the manuscript must state the version actually used.

[\[E5\] PFDelta author repository: dataset overview](#)

Released case families in the repository overview do not include a 2,224-bus GBnetwork.

[\[E6\] Park et al.: Compact Optimization Learning for AC Optimal Power Flow](#)

Primary abstract describes PCA-space learning, reconstruction and exact-solver warm starting.

[\[E7\] Microsoft Research: GridSFM technical white paper record](#)

Primary model record identifies multi-grid pretraining and released model variants.

[\[E8\] Li et al.: LUMINA primary preprint record](#)

Title, author and identifier spot-check; not an independent reproduction of its results.

[\[E9\] Puech et al.: GENCO primary preprint record](#)

Title, author and identifier spot-check.

[\[E10\] Chzhen and Donti: Improving Feasibility via Fast Autoencoder-Based Projections](#)

Primary record for the nonlinear learned-projection comparison.

Artifact and computation checks

The review checked all six rendered source pages, measured Table 3's embedded type size, compared Figure 2 visually with its accompanying text, and verified Eq. (5) by derivation plus 100 randomized numerical consistency checks. The numerical checks test the identity, not the authors' implementation. The toy Jacobian example is explicitly illustrative.

The manuscript repository URL returned HTTP 404 during a direct retrieval. This is an access result, not a conclusion about whether private code exists. No model training, full bibliography audit, benchmark replication or data-leakage determination was performed.